Matrixdock XAGM Audit

audited by Asymptotic

March 18, 2026

Summary

Asymptotic conducted a security audit of Matrixdock XAGM contract. The audit identified 1 low-severity issue. The issue has been remediated.

Audited code

We audited the following directories at the given commit:

- `xagm-sui/packages/messenger_lz`
- `xagm-sui/packages/minter`
- `xagm-sui/packages/mtoken`
- `xagm-sui/packages/xagm`

Legend

Issue severity

- **Critical** — Vulnerabilities which allow account takeovers or stealing significant funds, along with being easy to exploit.
- **High** — Vulnerabilities which either can have significant impact but are hard to exploit, or are easy to exploit but have more limited impact.
- **Medium** — Moderate risks with notable but limited impact.
- **Low** — Minor issues with minimal security implications.
- **Advisory** — Informational findings for security/code improvements.

Legal Disclaimer

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Scope

This security audit report focuses specifically on reviewing the Move smart contract code. The audit does not analyze or make any claims about other components of the system, including but not limited to:

- Frontend applications and user interfaces
- Backend services and infrastructure
- Off-chain components and integrations
- Deployment procedures and operational security
- Third-party dependencies and external services

Limitations

The findings and recommendations in this report are limited to the Move code implementation and its immediate interactions within the Sui blockchain environment.

Issues

L-1: Missing fee rate initialized check in `oz_per_token`

Severity: Low

Description

`oz_per_token` should check that `assert!(state.oz_per_token_base_time > 0, EAnnualFeeRateNotIn`
it shouldn't be callable before the annual fee rate is initialized.

Recommendation

Remediation

✓ **Remediated**

About the Auditor

Asymptotic provides a white-glove, formal-verification-based auditing service for Sui smart contracts.

Asymptotic Team

Audit Team

Lead Auditor

Andrei Stefanescu — Chief Scientist

- Led verification of AWS cryptographic algorithms using SAW at Galois
- Created first Ethereum smart contract verification tool
- Three-time International Math Olympiad silver medalist
- PhD in Computer Science from UIUC with David J. Kuck Outstanding Thesis Award
- ACM SIGPLAN Distinguished Paper Award at OOPSLA 2016
- Pioneered K Framework for programming language semantics and verification